

Q15 Using Rayleigh Criterion,

min angle of resolution between the two pixels, $\theta \approx \frac{\lambda}{b} = \frac{550 \times 10^{-9}}{2.0 \times 10^{-3}} = 2.75 \times 10^{-4} \text{ rad}$

$$\tan\left(\frac{\theta}{2}\right) = \frac{x/2}{D}$$

$$\begin{aligned}x &= 2D \tan\left(\frac{\theta}{2}\right) = 2(0.76) \tan\left(\frac{2.75 \times 10^{-4}}{2}\right) \\&= 2.1 \times 10^{-4} \text{ m} \\&= 0.21 \text{ mm}\end{aligned}$$

Answer: B

Q16 At equilibrium